Noregs miljø- og biovitskaplege universitet

Institutt for datavitskap

Digitalisering på Ås

INF205
Ressurseffektiv programmering

3 Datastrukturar

3.1 Grafdatastrukturar

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3.1 Grafdatastrukturar

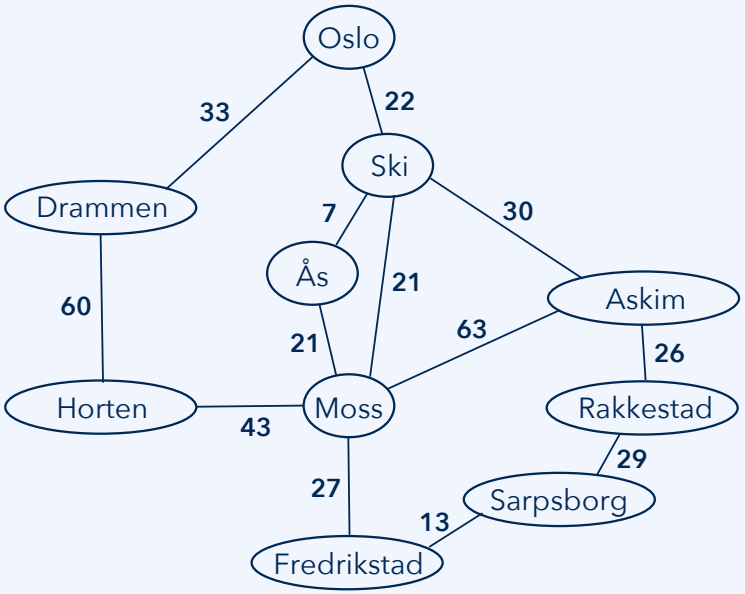
INF205

27. februar 2026

Ordliste

Adjacency list	**Naboliste**
Binary search tree	Binært søkjetre
Directed graph	Retta graf
Graph	Graf
Knowledge graph	Kunnskapsgraf
Tree	Tre

Graphs as non-sequential linked data structures



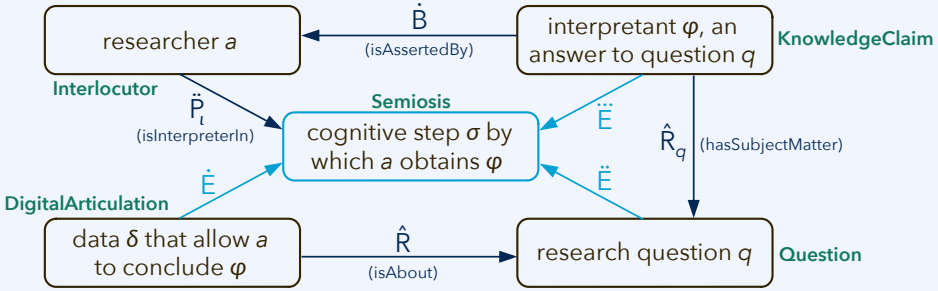
Sequential data structures arrange their items in a linear shape. Sometimes that is not the best solution, or it is not appropriate at all.

Linked data structures with a non-sequential shape are **graphs**, which includes the important special case of tree data structures.

A graph G = (V, E) is defined by its **nodes V**, which are also called vertices, and **edges E** that connect one node to another. Nodes and edges can be *labelled* to give the graph a meaning.

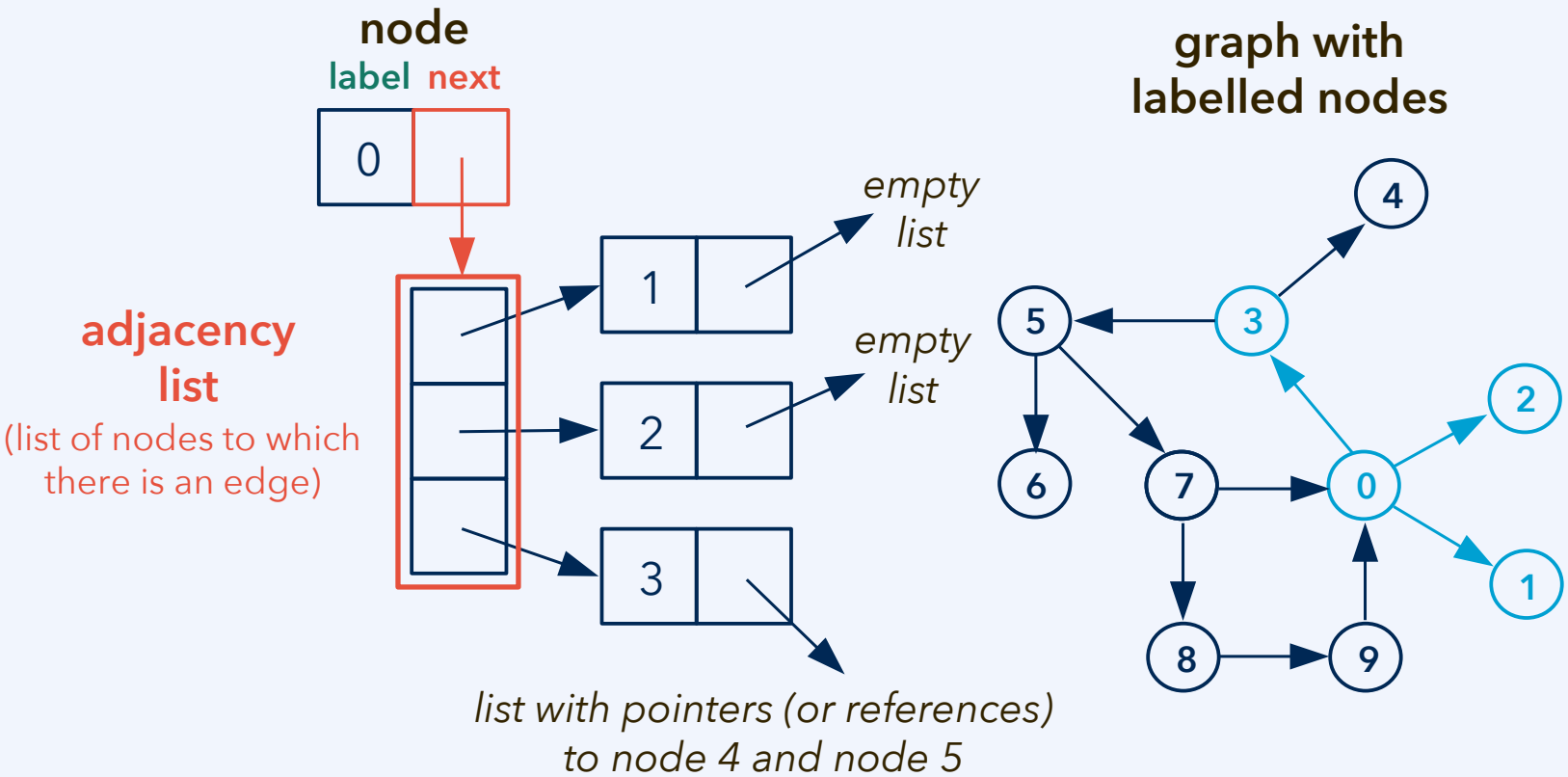
Graphs can be used to represent relations between objects, such as distances on a map, or as a **knowledge graph**.

Trees are often used as sorted data structures, for efficiency reasons.



Implementation: Adjacency lists (nabolister)

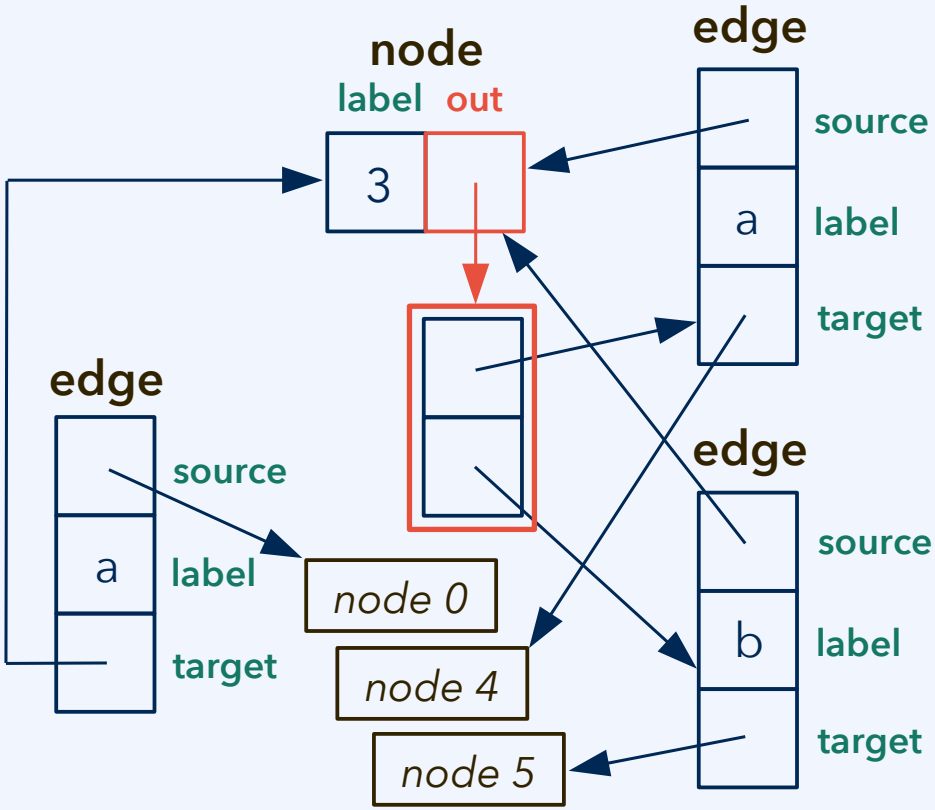
In a graph, one node can be connected to multiple other nodes. An **adjacency list** (with various possible implementations) can be used to manage these links.

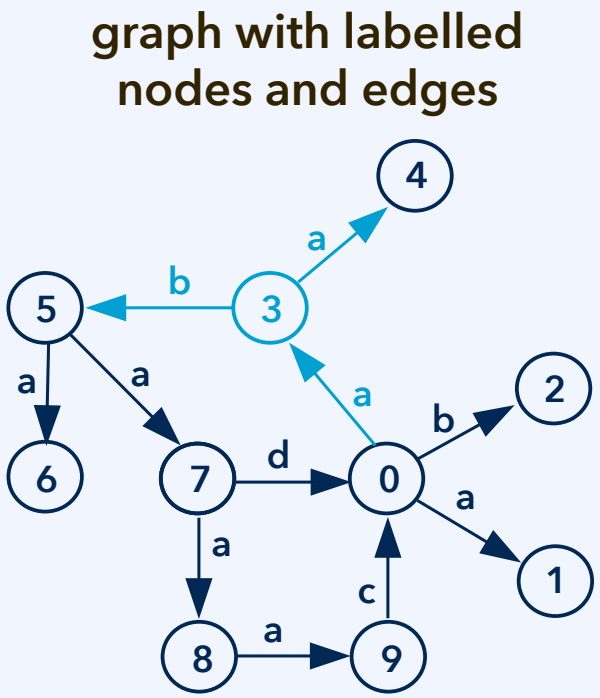


Doubly-linked version of this: Two lists, for incoming and for outgoing edges.

Implementation: Incidence lists (insidenslister)

An **incidence list** is a list of edges to which a node is incident. For adjacency lists or incidence lists, various data structures can be used, *e.g.*, dynamic arrays.

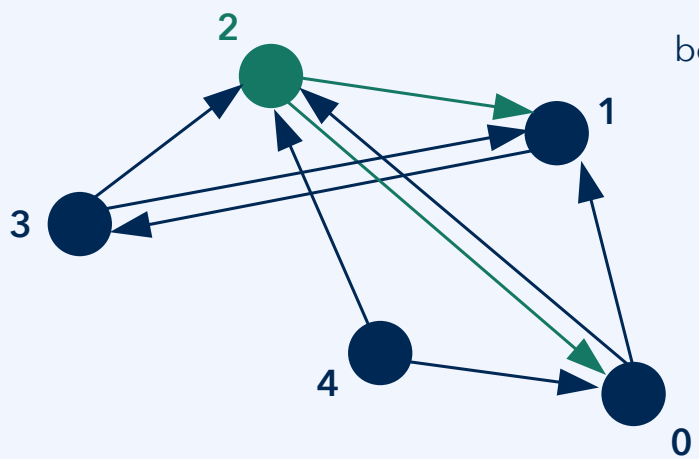




Doubly-linked version of this: Two lists, for incoming and for outgoing edges.

Implementation: Adjacency matrix (nabomatrise)

Matrix-like data structures include two-dimensional arrays, *i.e.*, arrays where the individual elements are accessed by double indexing. The most relevant use for graphs is the **adjacency matrix**. (Also possible: An incidence matrix.)

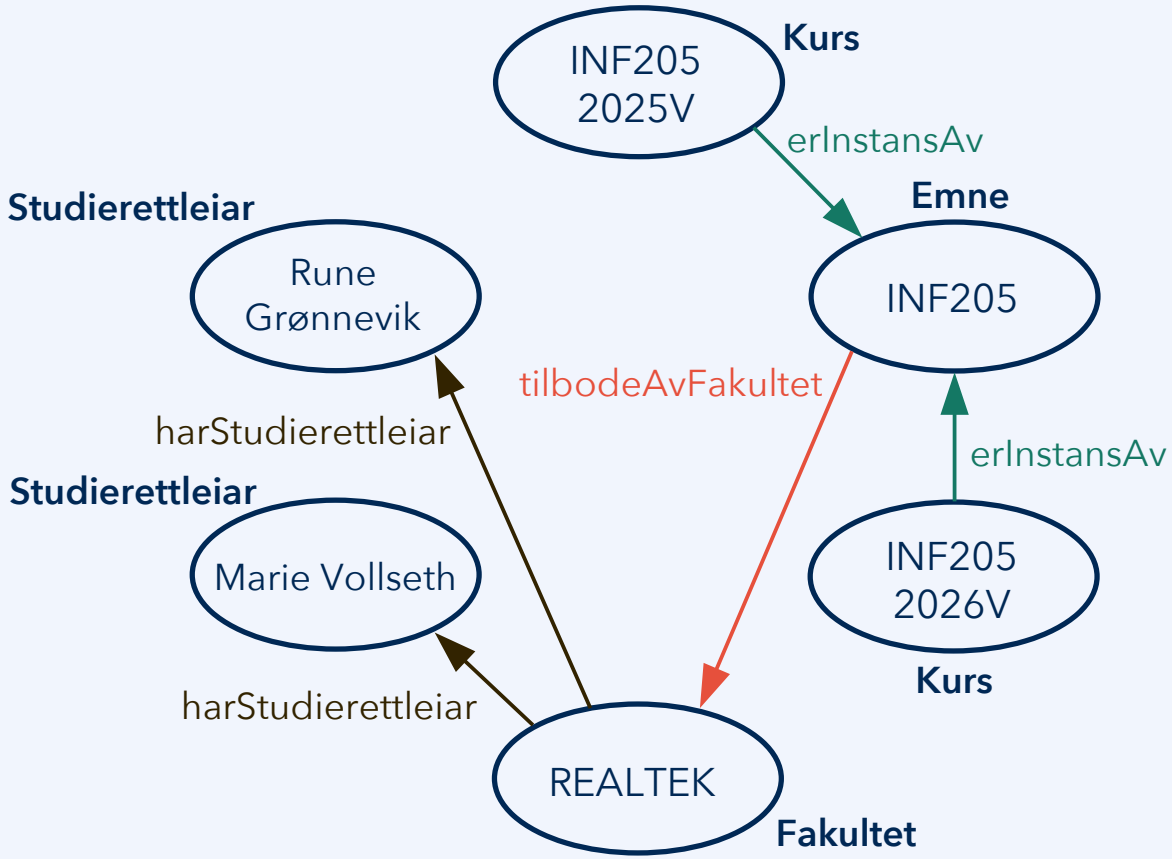



```
bool adj[5][5]={   {true, true,  true,  false, false},   out of node 0
                   {false,false, false, true,  false},   out of node 1
                   {true, true,  false, false, false},   out of node 2
                   {false,true,  true,  false, false},   out of node 3
                   {true, false, true,  false, false}  };  out of node 4
```

For a sparse graph, the vast majority of entries in the 2D array/matrix is “false”. Adjacency matrices are commonly only used when expecting a **dense graph**.

Semantics represented as a graph

Modern knowledge bases represent knowledge about the state of affairs as **knowledge graphs**. These graphs are understood as part of one **semantic web**.



Knowledge graphs contain **individuals** (objects) as nodes.

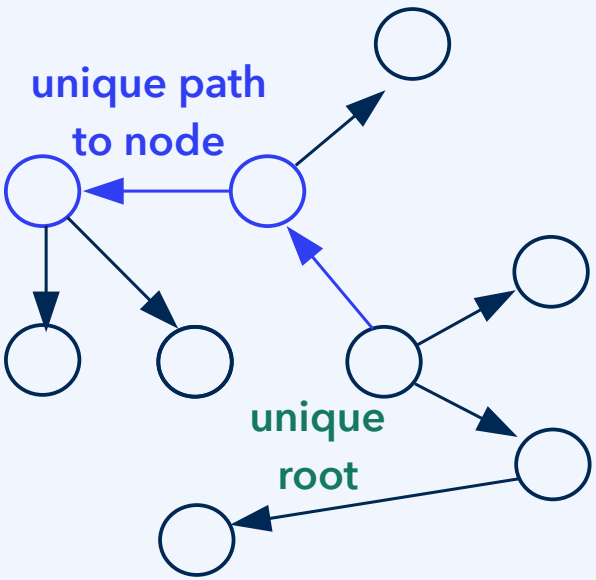
They contain **relations** (binary predicates) as edges.

They may also visually represent the instantiation of **concepts** (classes).

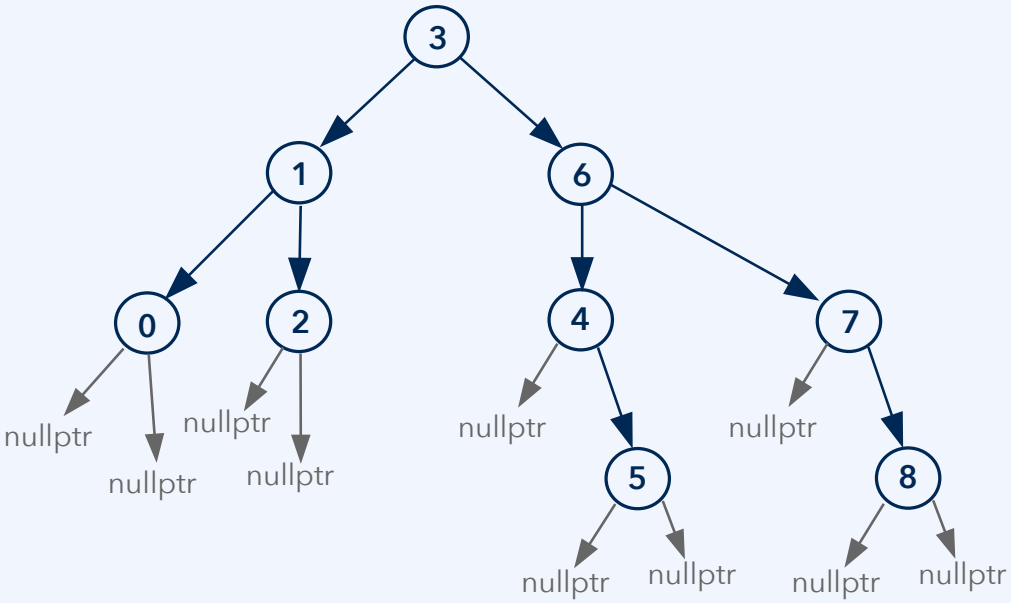
Tree data structures

Trees are a special kind of graph; or graphs are a generalization of trees:

tree (a kind of graph)



a binary search tree



Definition ("tree"; in the literature, also: "out-tree" or "rooted tree")

A tree is a graph with [a root and] a unique path from the root to each node.

Glossar

Graf

Kunnskapsgraf

Tre

Glossar

Graf

Definisjon: «samling [av] noder (punkter) og kanter (linjer) som forbinder noen, eller alle, av nodene» (T. Aambø i SNL om grafteori)

Ein graf kan dimed òg definerast som eit par $G = (V, E)$ av ei nodemengde V og ei kantemengde $E \subseteq V \times V$.

Kunnskapsgraf

Definisjon: graf der nodar er objekt og kantar er binære forhold mellom objekt

Tre

Definisjon: asyklisk samanhengande graf, dvs. graf der det ikkje finst syklar [òg «syklusar»], slik at det er éin sti mellom eitkvart par nodar

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